LESSON 8.5

Understand Inequalities and Their Solutions

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can graph the solutions of an inequality on a number line.

WHY THIS MATTERS Detective Okafor is building a timeline. The suspect was seen at the scene no earlier than 2:00 PM, meaning the arrival time t satisfies t ≥ 2. She needs to plot this on a timeline to share with her team. How should she mark the number line?

REMEMBER

Understand Inequalities and Their Solutions

Graphing Inequalities on the Number Line Graphing an inequality shows every value that makes it true. I draw a circle at the boundary number and shade the side that works. The circle tells me if the boundary is included.

1. Draw the boundary circle at the given constant number.
2. Open Circle (○) for < or > (boundary value is not included in solution set).
3. Closed Circle (●) for ≤ or ≥ (boundary value is included). Shade right for >, left for <

Words: **Number line *(Recta numérica)*** **Open circle *(Círculo abierto)*** **Closed circle *(Círculo cerrado)*** **Solution set *(Conjunto solución)*** **Inequality *(Desigualdad)***

1 **Circle the letter of the best answer. Show how you know.**

Which inequality represents: 'You must be at least 48 inches tall to ride'?

(A) h > 48 (B) h ≥ 48

(C) h < 48 (D) h ≤ 48

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which inequality represents: 'A store can hold no more than 75 people'?

(A) $p < 75$
(B) $p \geq 75$
(C) $p \leq 75$
(D) $p > 75$

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which inequality represents: 'The speed limit is more than 25 mph'?

(A) $s \geq 25$
(B) $s < 25$
(C) $s > 25$
(D) $s = 25$

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which inequality represents: 'The temperature is less than 40 degrees'?

(A) $t > 40$
(B) $t \leq 40$
(C) $t < 40$
(D) $t = 40$

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A detective determines that fewer than 12 people were at the scene. The inequality $p < 12$ represents the number of people p. She graphs this with an open circle at 12 and shading to the left. Why did the detective use an open circle at 12?

(A) Because 12 is negative

(B) Because the shading goes left

(C) Because 12 is not a whole number

(D) Because $p < 12$ does not include 12 itself

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A detective determines that fewer than 12 people were at the scene. The inequality $p < 12$ represents the number of people p. She graphs this with an open circle at 12 and shading to the left. Which number is a solution to $p < 12$?

(A) 11

(B) 12

(C) 13

(D) 15

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How do you decide between an open circle and a closed circle when graphing?

SENTENCE STARTER I use a ___ circle when the symbol is ___ because the boundary value ___ part of the solution.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 8.5

Understand Inequalities and Their Solutions

Set B · Version B · Second form · core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can graph the solutions of an inequality on a number line.

WHY THIS MATTERS Detective Okafor is building a timeline. The suspect was seen at the scene no earlier than 2:00 PM, meaning the arrival time t satisfies $t \geq 2$. She needs to plot this on a timeline to share with her team. How should she mark the number line?

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Detective Okafor is filing clues. Sort each inequality by the circle its graph uses.

Groups: Open circle (boundary NOT included) | Closed circle (boundary IS included)

_____ $c > 20$

_____ $t < 11$

_____ $w \geq 14$

_____ $m \leq 2$

2 **Write the letter of the matching item on each line.**

Match each clue inequality to the correct graph in the case file.

_____ $n > 13$

_____ $n \leq 16$

_____ $n \geq 20$

_____ $n < 17$

(A) Open circle at 13, shaded right

(B) Closed circle at 16, shaded left

(C) Closed circle at 20, shaded right

(D) Open circle at 17, shaded left

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A detective determines that fewer than 12 people were at the scene. The inequality $p < 12$ represents the number of people p. She graphs this with an open circle at 12 and shading to the left. How would the graph change if 12 or fewer people were at the scene?

(A) The shading would flip to the right

(B) Nothing would change

(C) The circle would be closed (filled in) at 12

(D) The circle would move to 11

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Clue: the witness ran more than 15 feet, so $d > 15$. On the number line, what do you draw?

(A) Open circle at 15, shade right

(B) Closed circle at 15, shade right

(C) Open circle at 15, shade left

(D) Closed circle at 15, shade left

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Clue: the getaway car held at most 4 suspects, so $s \leq 4$. How is this graphed?

(A) Closed circle at 4, shade left

(B) Open circle at 4, shade left

(C) Closed circle at 4, shade right

(D) Open circle at 4, shade right

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Level 2 Extension — Crack the Graphing Mistake

1. **Clue inequality** $k > 19$ (the loot weighed more than 19 pounds)
2. **Circle type** Closed circle at 19
3. **Shade direction** Shade right
4. **Final graph** Closed circle at 19, shaded right

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How do you decide between an open circle and a closed circle when graphing?

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